

SPECIAL TESTS — HIP JOINT

I. TESTS FOR HIP PATHOLOGY

1. HIP SCOUR TEST

PURPOSE

To assess for intra-articular hip pathology.

PROCEDURE

- Patient lies supine.
- Flex the hip and knee.
- Apply a gentle axial load through the femur.
- Move the hip through flexion/extension and abduction/adduction while maintaining compression.

POSITIVE SIGN

- Pain
- Catching
- Clicking
- Crepitus
- Restricted/abnormal movement

INDICATES

Possible intra-articular hip pathology, such as degenerative or labral pathology.

2. LEG ROLL TEST

PURPOSE

To assess for intra-articular hip pathology and restriction of passive hip rotation.

PROCEDURE

- Patient lies supine with the leg relaxed.
- Examiner gently rolls the leg from side to side, producing internal and external rotation of the hip.
- Compare with the opposite side.

POSITIVE SIGN

- Pain
- Markedly increased or decreased hip rotation
- Excessive external rotation compared with the opposite side

INDICATES

Possible hip joint pathology.

II. TEST FOR LABRAL LESION

4. ANTERIOR LABRAL TEAR TEST

PURPOSE

To assess for an acetabular labral lesion, particularly an anterior labral lesion.

PROCEDURE

- Patient lies supine.
- Flex the hip and bring it through the provocative combination of hip movement used to stress the anterior labrum.
- Reproduce the patient's familiar symptoms.
- Compare with the opposite side.

POSITIVE SIGN

- Reproduction of anterior hip/groin pain
- Clicking, catching or giving-way sensation may occur.

INDICATES

Possible acetabular labral lesion.

III. PEDIATRIC HIP PATHOLOGY

5. BARLOW'S TEST

PURPOSE

To determine whether an infant's hip is unstable and can be dislocated.

PROCEDURE

- Infant lies supine.
- Flex the hip and knee.
- Stabilize the pelvis.
- Gently adduct the hip.
- Apply gentle posteriorly directed pressure through the femur.

POSITIVE SIGN

- A palpable clunk as the femoral head dislocates.

INDICATES

Dislocatable/unstable hip, suggestive of developmental dysplasia of the hip (DDH).

6. GALEAZZI SIGN

PURPOSE

To detect apparent femoral shortening, particularly due to hip dislocation.

PROCEDURE

- Infant lies supine.
- Flex both hips and knees to approximately 90°.
- Place the feet together.
- Compare the height of both knees.

POSITIVE SIGN

- One knee is lower than the other.

INDICATES

- Developmental hip dislocation/dysplasia
- Femoral shortening
- Other limb-length discrepancy

7. ORTOLANI SIGN

PURPOSE

To detect a dislocated hip that can be reduced.

PROCEDURE

- Infant lies supine with hips and knees flexed.
- Stabilize the pelvis.
- Gently abduct the hip while applying pressure over the greater trochanter in an appropriate direction to reduce the femoral head.

POSITIVE SIGN

- A palpable “clunk” as the femoral head reduces into the acetabulum.

INDICATES

Reducible hip dislocation → DDH.

★ Barlow vs Ortolani

Barlow → OUT = dislocates

Ortolani → IN = reduces

IV. TESTS FOR MUSCLE DYSFUNCTION

8. HIP LAG SIGN

PURPOSE

To assess hip extensor muscle function, particularly the ability to actively maintain hip extension.

PROCEDURE

- Patient lies prone.
- Examiner passively extends the hip.
- Ask the patient to maintain the hip in the extended position after the examiner releases support.

POSITIVE SIGN

- The patient cannot maintain the extended position.
- The hip falls back toward flexion.

INDICATES

Hip extensor weakness/dysfunction.

10. THOMAS TEST

PURPOSE

To assess hip flexor tightness/hip flexion contracture.

PROCEDURE

- Patient lies supine near the edge of the table.
- Patient flexes one hip and knee toward the chest to flatten the lumbar spine.
- Observe the position of the opposite leg.

POSITIVE SIGN

- Opposite thigh rises from the table → hip flexor tightness.
- Knee extends → possible rectus femoris tightness.
- Hip abducts → possible TFL/ITB tightness.

INDICATES

Hip flexor muscle tightness/contracture.

V. TESTS FOR HIP DISPLACEMENT

11. NELATON'S LINE

PURPOSE

To assess the position of the greater trochanter in relation to the pelvis.

PROCEDURE

- Patient lies in the appropriate position with the hip examined.
- Draw an imaginary line from the ASIS to the ischial tuberosity.
- Assess the position of the greater trochanter relative to this line.

POSITIVE FINDING

- Greater trochanter lies above Nelaton's line.

INDICATES

Possible superior displacement of the femur/hip dislocation.

12. BRYANT'S TRIANGLE

PURPOSE

To detect proximal displacement of the greater trochanter and apparent shortening of the femur.

PROCEDURE

- Patient lies supine.
- From the ASIS, draw a perpendicular line to the bed.
- Draw a line from the greater trochanter to meet this perpendicular.
- Compare the triangles on both sides.

POSITIVE FINDING

- Shortening of the base of Bryant's triangle on the affected side.

INDICATES

Possible hip dislocation or proximal displacement of the femur.

VI. SACROILIAC JOINT TESTS

14. GAENSLEN'S TEST

PURPOSE

To assess SI joint pathology.

PROCEDURE

- Patient lies near the edge of the table.
- One hip is flexed toward the chest.
- The opposite leg hangs over the edge.
- Apply gentle pressure to produce opposite pelvic movements.

POSITIVE SIGN

- Reproduction of pain around the SI joint.

INDICATES

Sacroiliac joint dysfunction/pathology.

VII. OTHER MUSCLE DYSFUNCTION TESTS

17. TRIPOD SIGN

PURPOSE

To assess hamstring tightness.

PROCEDURE

- Patient sits with the legs extended.
- Examiner observes the patient's sitting posture.
- Patient attempts to sit upright.

POSITIVE SIGN

- Patient leans backward and supports themselves with their hands.
- Knee flexion may occur because of hamstring tightness.

INDICATES

Hamstring tightness/shortness.

★ COMPLETE HIP REVISION TABLE

Category	Test	Indicates
Hip pathology	Hip Scour	Intra-articular hip pathology
Hip pathology	Leg Roll	Hip joint pathology
Hip pathology	Patrick's/FABER	Hip/SI pathology
Labral lesion	Anterior Labral Tear Test	Acetabular labral lesion
Pediatric hip	Barlow's	Dislocatable/unstable hip
Pediatric hip	Galeazzi	Hip dislocation/femoral shortening
Pediatric hip	Ortolani	Reducible hip dislocation
Muscle dysfunction	Hip Lag Sign	Hip extensor weakness
Muscle dysfunction	90–90 SLR	Hamstring tightness
Muscle dysfunction	Thomas Test	Hip flexor tightness
Muscle dysfunction	Tripod Sign	Hamstring tightness
Muscle dysfunction	Trendelenburg Sign	Hip abductor weakness
Hip displacement	Nelaton's Line	Greater trochanter displacement
Hip displacement	Bryant's Triangle	Hip displacement/shortening
SI joint	FABER	Hip/SI pathology
SI joint	Gaenslen's	SI joint pathology
SI joint	Gillet's	SI joint mobility
SI joint	March Test	SI joint dysfunction